

B4 Nuclear and Particle Physics, 2021 — Model Solutions

Question 1 — Conservation laws, e^+e^- collisions and the Z^0

Problem. Discuss colour and flavour conservation; explain $e^+e^- \rightarrow \mu^+e^- + \text{missing}$; find the two-EM-shower fraction of $\tau\tau$ at a $\mu^+\mu^-$ collider; ratio of e^+e^- to hadronic interactions just below the Z^0 peak.

(a) Colour and flavour conservation

Colour. $SU(3)_C$ exact gauge symmetry: every QCD vertex conserves colour; γ , $W^\pm$, Z^0 are colour singlets, so EM and weak vertices also conserve colour. Evidence: $R \equiv \sigma(e^+e^- \rightarrow \text{had})/\sigma(e^+e^- \rightarrow \mu^+\mu^-) = N_c \sum_q e_q^2$ requires $N_c = 3$; $\Gamma(\pi^0 \rightarrow \gamma\gamma) \propto N_c^2$; three jets in $e^+e^- \rightarrow q\bar{q}g$.

Quark flavour. Strong + EM diagonal in flavour (vertices conserve S, C, B, T, I_3). Charged current $W^\pm$ violates flavour via CKM ($\Delta S = \pm 1$ at one vertex); neutral current Z^0 is flavour-diagonal at tree level (GIM). Evidence: $K^0 \leftrightarrow \bar{K}^0$ oscillations, $K^+ \rightarrow \mu^+\nu$, $B \rightarrow D\ell\nu$; suppression of $K_L \rightarrow \mu^+\mu^-$ confirms GIM.

Lepton flavour. L_e, L_μ, L_τ each conserved at all vertices in the SM (massless- ν limit); broken only by neutrino oscillation in propagation. Evidence: $\text{BR}(\mu \rightarrow e\gamma) < 10^{-13}$; success of $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$.

(b) Two SM channels for $e^+e^- \rightarrow \mu^+e^- + \text{missing}$

Visible $L_e = +1$, $L_\mu = -1$, $Q = 0$: missing must carry $L_e = -1$, $L_\mu = +1$.

Channel 1: W -pair production, $\sqrt{s} \gtrsim 2M_W \simeq 161 \text{ GeV}$:

$$e^+e^- \rightarrow Z^0/\gamma^* \rightarrow W^+W^-, \quad W^+ \rightarrow \mu^+\nu_\mu, \quad W^- \rightarrow e^-\bar{\nu}_e.$$

Branching $\text{BR}(W \rightarrow \ell\nu) \simeq 1/9$ per flavour; rate $\propto (1/9)^2$ of σ_{WW} .

Channel 2: τ -pair production, $\sqrt{s} \geq 2m_\tau \simeq 3.5 \text{ GeV}$ (rate maximised at the Z^0 pole):

$$e^+e^- \rightarrow \gamma^*/Z^0 \rightarrow \tau^+\tau^-, \quad \tau^+ \rightarrow \mu^+\nu_\mu\bar{\nu}_\tau, \quad \tau^- \rightarrow e^-\bar{\nu}_e\nu_\tau.$$

Feynman diagram (Channel 1). e^+e^- annihilate at a vertex into s -channel γ^*/Z^0 which couples to W^+W^- at a second vertex; $W^+ \rightarrow \mu^+\nu_\mu$ and $W^- \rightarrow e^-\bar{\nu}_e$ at the two leptonic vertices.

Feynman diagram (Channel 2). e^+e^- annihilate via s -channel γ^*/Z^0 into $\tau^+\tau^-$; each τ decays via its own W propagator: $\tau^+ \rightarrow \bar{\nu}_\tau W^+$, $W^+ \rightarrow \mu^+\nu_\mu$; $\tau^- \rightarrow \nu_\tau W^-$, $W^- \rightarrow e^-\bar{\nu}_e$.

Missing-energy sketches.

Channel 1: each W decay is two-body, $E_\ell^* = M_W/2$ in the W rest frame (isotropic). Boost to lab ($E_W = \sqrt{s}/2$, $p_W = \sqrt{s/4 - M_W^2}$) flattens the spectrum:

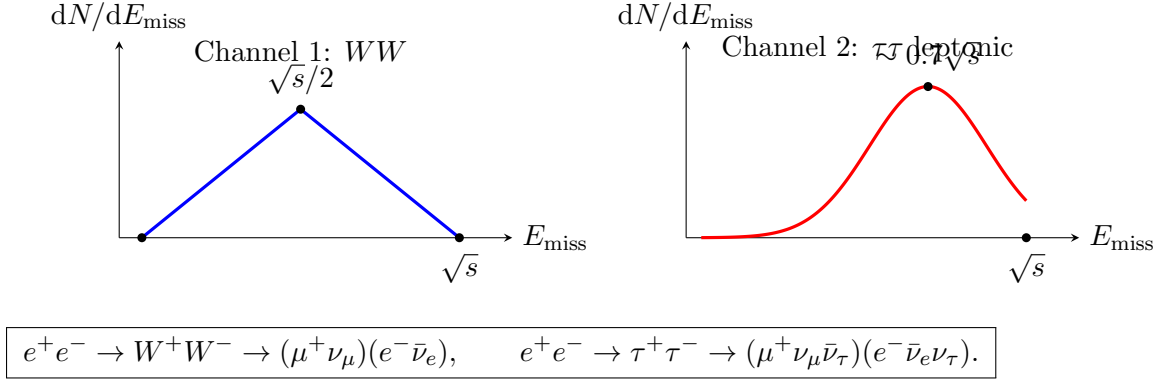
$$E_\ell^{\text{lab}} \in \left[\frac{1}{2}(E_W - p_W), \frac{1}{2}(E_W + p_W) \right].$$

Two flat (box) charged-lepton spectra convolve to a triangular missing-energy distribution centred at $\sqrt{s}/2$.

Channel 2: leptonic τ decay has the Michel spectrum, charged-lepton energy

$$\frac{1}{\Gamma} \frac{d\Gamma}{dx} = 2x^2(3-2x), \quad x = 2E_\ell/m_\tau,$$

peaked at $x = 1$. Boosting to lab gives a softer charged-lepton spectrum, hence a missing-energy distribution that is broad and unimodal, peaking near $\sim 0.7\sqrt{s}$.



(c) Two-EM-shower τ -pair fraction

At $\sqrt{s} = M_Z$, $Z^0 \rightarrow \tau^+\tau^-$ produces back-to-back taus. “Two high-energy EM showers fully contained in the EMCAL”, with no muon track and no hadronic activity, requires both taus to decay leptonically to electrons:

$$\tau \rightarrow e \bar{\nu}_e \nu_\tau, \quad \text{BR} \simeq 17.8\%.$$

Both legs:

$$P_{ee} = [\text{BR}(\tau \rightarrow e\nu\bar{\nu})]^2 = (0.178)^2.$$

Evaluate:

$$P_{ee} \simeq 3.2\%. \quad (1)$$

(d) e^+e^- /hadrons a few MeV below the Z^0 peak

On the Z^0 resonance the cross-section ratio reduces to a partial-width ratio:

$$\frac{\sigma(e^+e^- \rightarrow e^+e^-)}{\sigma(e^+e^- \rightarrow \text{had})} = \frac{\Gamma_{ee}}{\Gamma_{\text{had}}}.$$

Lepton universality: $\Gamma_{ee} = \Gamma_{\mu\mu} = \Gamma_{\tau\tau} \simeq 83.9 \text{ MeV}$; PDG $\Gamma_{\text{had}} \simeq 1744 \text{ MeV}$. Substitute:

$$\frac{\Gamma_{ee}}{\Gamma_{\text{had}}} = \frac{83.9}{1744}.$$

Evaluate:

$$\frac{N(e^+e^-)}{N(\text{had})} \simeq 4.8\%. \quad (2)$$

Equivalently $R_Z \equiv \Gamma_{\text{had}}/\Gamma_{\ell\ell} \simeq 20.8$, consistent with five accessible quark flavours $\times N_c = 3$ weighted by electroweak couplings.

Question 2 — Bubble-chamber identification and the $\Delta(1232)$

Problem. List conserved quantum numbers; identify charged tracks; deduce neutrals X, Y in $\pi^- p$ event; mass and lifetime of the $\pi^+ p$ resonance; non-relativistic Breit–Wigner shape and deviations.

(a) Conserved quantum numbers

Quantity	S	EM	W	Comment
$E, \mathbf{p}, \mathbf{J}$	✓	✓	✓	spacetime symmetries
Charge Q	✓	✓	✓	$U(1)_{\text{EM}}$
Colour	✓	✓	✓	$SU(3)_C$ exact
Baryon B	✓	✓	✓	accidental global $U(1)$
Lepton family L_ℓ	✓	✓	✓*	*broken by ν oscillations
Quark flavour S, C, B, T	✓	✓	×	CKM at W vertex
P	✓	✓	×	maximal P violation in W
C	✓	✓	×	
CP/T	✓	✓	×	CKM, PMNS phases
Isospin I	✓ [†]	×	×	[†] $m_u \neq m_d$
G -parity	✓ [†]	×	×	approximate

(b) Identification of charged tracks

Magnetic rigidity $p = qBR$ from track curvature gives momentum (sign $\Rightarrow$ charge sign). Calorimetric energy E at terminus combined with p gives $m^2 = E^2 - p^2 c^2$, hence species. dE/dx ionisation, time-of-flight, or Cherenkov supplement when $\beta \rightarrow 1$.

(c) Identification of X and Y

Initial state: $\pi^- p$ has $Q = 0$, $B = +1$, $S = 0$, $L_\ell = 0$.

Vertex producing $\pi^+ \pi^-$ (downstream). Neutral parent ($B = 0$) decays to $\pi^+ \pi^-$. Strangeness-zero candidates (ρ^0 , ω , f^0) decay strongly with $c\tau \ll 1$ fm, no visible flight. Visible vee in a chamber requires weak decay: $K_S^0 \rightarrow \pi^+ \pi^-$ (BR 69%, $c\tau = 2.68$ cm). Strangeness $|S| = 1$.

$$X = K_S^0.$$

Vertex producing $p\pi^-$. Neutral with $B = +1$, decaying to $p + \pi^-$. Candidates: $\Lambda(uds) \rightarrow p\pi^-$ ($c\tau = 7.89$ cm) or $\Sigma^0 \rightarrow \Lambda\gamma$. The hint $\Sigma^0 \rightarrow \Lambda\gamma$ is electromagnetic with $\tau \sim 7 \times 10^{-20}$ s, $c\tau \sim 2 \times 10^{-11}$ m: any Σ^0 decays before leaving a track, so the visible neutral track is the Λ itself (whether produced directly or via $\Sigma^0 \rightarrow \Lambda\gamma$).

Strangeness conservation in strong production: X carries $S = +1$, so Y must carry $S = -1$. Both Λ and Σ^0 have $S = -1$. Visible flight pattern selects Λ as the chamber-traversing neutral:

$$Y = \Lambda.$$

Production reaction (associated production):

$$\pi^- + p \rightarrow \Lambda + K_S^0, \quad \Lambda \rightarrow p\pi^-, \quad K_S^0 \rightarrow \pi^+ \pi^-.$$

(d) Mass and lifetime of the π^+p resonance

Read off graph: peak at $T_\pi^{\text{peak}} \simeq 195 \text{ MeV}$, FWHM $\Delta T_\pi \simeq 180 \text{ MeV}$.

Invariant mass squared, fixed-target (p at rest):

$$s = m_\pi^2 + m_p^2 + 2m_p E_\pi^{\text{lab}}, \quad E_\pi^{\text{lab}} = T_\pi + m_\pi. \quad (3)$$

Substitute $m_\pi = 139.6 \text{ MeV}$, $m_p = 938.3 \text{ MeV}$, $T_\pi = 195 \text{ MeV}$:

$$E_\pi^{\text{lab}} = 195 + 139.6 = 334.6 \text{ MeV}.$$

Insert into (3):

$$s = (0.1396)^2 + (0.9383)^2 + 2(0.9383)(0.3346) \text{ GeV}^2.$$

Evaluate:

$$s = 0.0195 + 0.8804 + 0.6280 = 1.528 \text{ GeV}^2.$$

Take the square root:

$$\sqrt{s} = 1.236 \text{ GeV}.$$

$$\boxed{M_R \simeq 1.23 \text{ GeV}/c^2 \quad (\Delta(1232)).} \quad (4)$$

Width transform: differentiate (3) at fixed m_π, m_p :

$$2\sqrt{s} \Delta\sqrt{s} = 2m_p \Delta T_\pi.$$

Solve for $\Gamma \simeq \Delta\sqrt{s}$:

$$\Gamma = \frac{m_p}{\sqrt{s}} \Delta T_\pi.$$

Substitute:

$$\Gamma = \frac{0.938}{1.236} \times 180 \text{ MeV} = 137 \text{ MeV}.$$

Lifetime $\tau = \hbar/\Gamma$:

$$\tau = \frac{6.58 \times 10^{-22} \text{ MeV s}}{137 \text{ MeV}}.$$

Evaluate:

$$\boxed{\Gamma \simeq 137 \text{ MeV}, \quad \tau \simeq 4.8 \times 10^{-24} \text{ s}.} \quad (5)$$

Strong-decay timescale, consistent with PDG ($\Gamma = 117 \text{ MeV}$).

(e) Non-relativistic Breit–Wigner and deviations

Single isolated partial wave, projectile spins $s_1 = 0$ (π), $s_2 = 1/2$ (p), $J = 3/2$ (Δ):

$$\sigma_{\text{el}}(E) = \frac{4\pi}{k^2} \frac{2J+1}{(2s_1+1)(2s_2+1)} \frac{\Gamma^2/4}{(E-E_R)^2 + \Gamma^2/4}. \quad (6)$$

Spin factor:

$$\frac{2J+1}{(2s_1+1)(2s_2+1)} = \frac{4}{1 \cdot 2} = 2.$$

On peak:

$$\sigma_{\text{el}}^{\text{peak}} = \frac{8\pi}{k_R^2}.$$

CM wavenumber from (3):

$$k_R^2 = \frac{[s - (m_\pi + m_p)^2][s - (m_\pi - m_p)^2]}{4s}.$$

Substitute $s = 1.528 \text{ GeV}^2$, $(m_\pi + m_p)^2 = 1.162$, $(m_\pi - m_p)^2 = 0.638 \text{ GeV}^2$:

$$k_R^2 = \frac{(1.528 - 1.162)(1.528 - 0.638)}{4 \times 1.528} \text{ GeV}^2 = \frac{0.366 \times 0.890}{6.112} \text{ GeV}^2.$$

Evaluate:

$$k_R^2 = 0.0533 \text{ GeV}^2 \Rightarrow k_R \simeq 231 \text{ MeV}.$$

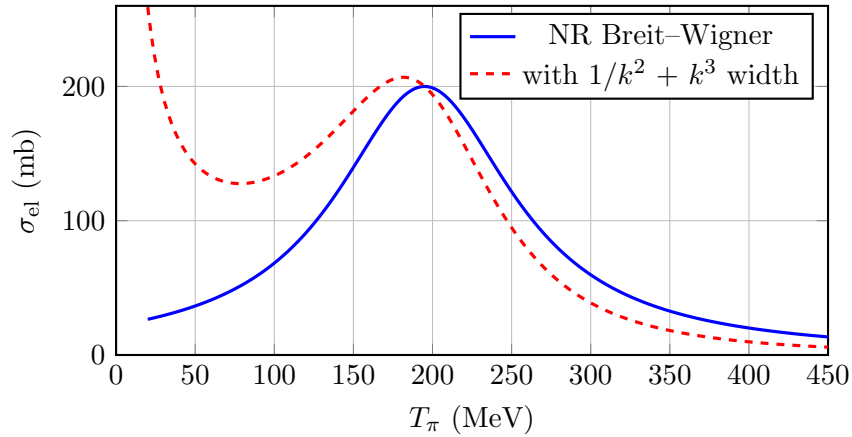
Peak cross-section using $1 \text{ GeV}^{-2} = 0.389 \text{ mb}$:

$$\sigma_{\text{el}}^{\text{peak}} = \frac{8\pi}{0.0533} \text{ GeV}^{-2} = 471 \text{ GeV}^{-2} = 471 \times 0.389 \text{ mb}.$$

$$\sigma_{\text{el}}^{\text{peak}} \simeq 184 \text{ mb},$$

(7)

matching the observed peak ($\sim 200 \text{ mb}$) within $\sim 10\%$.



Deviations from (6).

1. Prefactor $1/k^2$ skews the peak towards lower energies.
2. Energy-dependent P -wave partial width: $\Delta(J^P = 3/2^+) \rightarrow \pi N$ has $L = 1$, hence $\Gamma_{\pi N}(E) \propto k^{2L+1} = k^3$. Phase-space suppression at low T_π , broadening at high T_π .
3. Non-resonant S -wave background and (above $\sim 400 \text{ MeV}$) onset of inelastic channels and the next resonance ($N^*(1440)$).

The asymmetric shape (steep rise, slow fall) is the signature of the P_{33} ($I = 3/2, J = 3/2, L = 1$) state.

Question 3 — Born approximation and the deuteron

Problem. Define the Born scattering amplitude; evaluate it for a Gaussian potential $U(r) = U_0 e^{-r^2/2\sigma^2}$; compute σ_{tot} ; estimate intrinsic nucleon momentum at $\sigma \sim 1 \text{ fm}$, deuteron U_0 from $|E| = 2.2 \text{ MeV}$, and mean free path of 10 MeV neutrons in water.

(a) Born amplitude and validity

Asymptotic wavefunction: incoming plane wave plus outgoing spherical wave,

$$\psi(\mathbf{r}) \xrightarrow{r \rightarrow \infty} e^{i\mathbf{k} \cdot \mathbf{r}} + f(\theta, \phi) \frac{e^{ikr}}{r}.$$

Iterating the Lippmann–Schwinger equation once with unperturbed plane waves:

$$f^{\text{Born}}(\theta, \phi) = -\frac{m}{2\pi\hbar^2} \int e^{-i\mathbf{q} \cdot \mathbf{r}} U(\mathbf{r}) d^3\mathbf{r}, \quad \mathbf{q} = \mathbf{k}' - \mathbf{k}. \quad (8)$$

Validity. Scattered wave small inside the potential range, $|U|a \ll \hbar v$ at high k , $|U| \ll \hbar^2/(ma^2)$ at low k ; fails near bound states or resonances. (Convention below: U has units of energy, $f = -(m/2\pi\hbar^2)\tilde{U}(\mathbf{q})$.)

(b) Forward Born amplitude for a Gaussian

Elastic: $|\mathbf{k}| = |\mathbf{k}'| = k$, momentum transfer

$$q^2 = |\mathbf{k} - \mathbf{k}'|^2 = 2k^2(1 - \cos\theta).$$

Fourier transform of the Gaussian (factorise into three 1D integrals):

$$\tilde{U}(\mathbf{q}) = \int e^{-i\mathbf{q} \cdot \mathbf{r}} U_0 e^{-r^2/2\sigma^2} d^3\mathbf{r} = U_0 (2\pi\sigma^2)^{3/2} e^{-q^2\sigma^2/2}.$$

Insert into (8):

$$f(\theta, k) = -\frac{m}{2\pi\hbar^2} U_0 (2\pi\sigma^2)^{3/2} e^{-q^2\sigma^2/2}.$$

Substitute $q^2 = 2k^2(1 - \cos\theta)$:

$$f(\theta, k) = -\frac{m U_0 (2\pi)^{3/2} \sigma^3}{2\pi\hbar^2} e^{-\sigma^2 k^2 (1 - \cos\theta)}.$$

Tidy the prefactor:

$$f(\theta, k) = A e^{-\sigma^2 k^2 (1 - \cos\theta)}, \quad A = -\frac{m U_0 \sigma^3 \sqrt{2\pi}}{\hbar^2}. \quad (9)$$

(c) Total cross-section

Integrate $|f|^2$ over solid angle:

$$\sigma_{\text{tot}}(k) = \int |f|^2 d\Omega = 2\pi A^2 \int_0^\pi e^{-2\sigma^2 k^2 (1 - \cos\theta)} \sin\theta d\theta.$$

Substitute $u = 1 - \cos\theta$, $du = \sin\theta d\theta$, $u \in [0, 2]$:

$$\sigma_{\text{tot}}(k) = 2\pi A^2 \int_0^2 e^{-2\sigma^2 k^2 u} du.$$

Evaluate the integral:

$$\sigma_{\text{tot}}(k) = 2\pi A^2 \frac{1 - e^{-4\sigma^2 k^2}}{2\sigma^2 k^2}.$$

Tidy:

$$\sigma_{\text{tot}}(k) = B(1 - e^{-4\sigma^2 k^2}), \quad B = \frac{\pi A^2}{\sigma^2 k^2} = \frac{2\pi^2 m^2 U_0^2 \sigma^4}{\hbar^4 k^2}. \quad (10)$$

Limits: $\sigma k \ll 1 \Rightarrow \sigma_{\text{tot}} \rightarrow 4\pi A^2$ (constant, S -wave); $\sigma k \gg 1 \Rightarrow \sigma_{\text{tot}} \rightarrow B \propto 1/k^2$.

(d) Intrinsic nucleon momentum at $\sigma \sim 1$ fm

Heisenberg, $\Delta r \sim \sigma$:

$$p \sim \frac{\hbar}{\sigma} = \frac{197 \text{ MeV fm}}{1 \text{ fm}}.$$

$p \sim 200 \text{ MeV}/c.$

(11)

Non-relativistic kinetic energy:

$$T \sim \frac{p^2}{2m_N} = \frac{(200)^2}{2(940)} \text{ MeV} \simeq 21 \text{ MeV}.$$

(e) Estimate of U_0 from $|E| = 2.2$ MeV

Two-body deuteron: reduced mass $\mu = m_p m_n / (m_p + m_n) \simeq m_N / 2 \simeq 470 \text{ MeV}/c^2$. Variational ansatz of width σ :

$$E \approx \frac{\hbar^2}{2\mu\sigma^2} + U_0 \langle e^{-r^2/2\sigma^2} \rangle.$$

Take $\langle e^{-r^2/2\sigma^2} \rangle \sim 1$ for an order-of-magnitude estimate. Set $E = -2.2 \text{ MeV}$:

$$\frac{(\hbar c)^2}{2\mu c^2 \sigma^2} + U_0 = -2.2 \text{ MeV}.$$

Kinetic term:

$$T = \frac{(197)^2}{2(470)(1)^2} \text{ MeV} \simeq 41 \text{ MeV}.$$

Solve for U_0 :

$$U_0 = -T - |E| = -41 - 2.2 \text{ MeV}.$$

$U_0 \simeq -43 \text{ MeV (attractive)}.$

(12)

Canonical depth of the nucleon–nucleon well; binding is the small difference of two $\sim 40 \text{ MeV}$ terms.

(f) Mean free path of a 10 MeV neutron in water

CM wavenumber for $E = 10 \text{ MeV}$:

$$\hbar c k = \sqrt{2m_n c^2 \cdot E} = \sqrt{2(940)(10)} \text{ MeV} = 137 \text{ MeV}.$$

Hence $k = 137/197 \text{ fm}^{-1} = 0.696 \text{ fm}^{-1}$, $\sigma k = 0.696$, $4\sigma^2 k^2 = 1.94$:

$$1 - e^{-4\sigma^2 k^2} = 1 - e^{-1.94} = 0.856.$$

Compute B from (10) using natural units. Group the prefactor:

$$\frac{m_n c^2 |U_0| \sigma^2}{(\hbar c)^2} = \frac{(940)(43)(1)^2}{(197)^2}.$$

Evaluate:

$$\frac{m_n c^2 |U_0| \sigma^2}{(\hbar c)^2} = \frac{40\,420}{38\,809} = 1.041.$$

Hence

$$B = \frac{2\pi^2}{(\sigma k)^2} (1.041)^2 \sigma^2 = \frac{2\pi^2(1.084)}{0.484} \text{ fm}^2.$$

Evaluate:

$$B \simeq 44.2 \text{ fm}^2 = 0.442 \text{ barn}.$$

Total cross-section:

$$\sigma_{\text{tot}} = B \times 0.856 = 0.378 \text{ barn} = 3.78 \times 10^{-25} \text{ cm}^2.$$

Hydrogen number density in water ($\rho = 1 \text{ g/cm}^3$, $M = 18 \text{ g/mol}$, two H per molecule):

$$n_H = \frac{2N_A\rho}{M} = \frac{2(6.02 \times 10^{23})}{18} \text{ cm}^{-3} = 6.69 \times 10^{22} \text{ cm}^{-3}.$$

Mean free path:

$$\lambda = \frac{1}{n_H \sigma_{\text{tot}}} = \frac{1}{(6.69 \times 10^{22})(3.78 \times 10^{-25})} \text{ cm}.$$

Evaluate:

$$\boxed{\lambda \simeq 40 \text{ cm.}} \quad (13)$$

Order-of-magnitude consistent with the moderation length in light water (Born underestimates by ~ 2 – 3 near a nearly-bound channel).

Question 4 — SEMF, fission and reactor neutrinos

Problem. Discuss the binding-energy-per-nucleon curve and the SEMF; estimate ΔE in the canonical ^{235}U fission; rate of $\bar{\nu}_e$ from $10 \text{ GW}_{\text{th}}$; detector mass at 200 km for $10 \text{ IBD events/month}$; explain factor-of-two suppression.

(a) Binding-energy curve and the SEMF

$$B(A, Z) = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(N-Z)^2}{A} + \delta(A, Z), \quad (14)$$

with $a_V \simeq 15.8$, $a_S \simeq 17.8$, $a_C \simeq 0.71$, $a_A \simeq 23.7 \text{ MeV}$; $\delta = \pm a_P A^{-1/2}$, $a_P \simeq 12 \text{ MeV}$.

Volume $+a_V A$. Saturation of the short-range ($\lesssim 2 \text{ fm}$) NN force: each nucleon binds only to neighbours, contribution $\propto A$.

Surface $-a_S A^{2/3}$. Surface nucleons miss neighbours; surface area $\propto R^2 \propto A^{2/3}$. Dominant for light nuclei, drives the rise of B/A from $A = 1$ to ~ 20 .

Coulomb $-a_C Z(Z-1)/A^{1/3}$. Self-energy of Z protons in radius $\propto A^{1/3}$. Drives the slow decline beyond $A \gtrsim 60$ and fission instability of the heaviest nuclei.

Asymmetry $-a_A (N-Z)^2/A$. Pauli exclusion penalty for $N \neq Z$; favours $N = Z$ light nuclei but is overcome by Coulomb in heavy nuclei (valley of stability bends to $N > Z$).

Pairing $\pm a_P A^{-1/2}$. Like-pair coupling to $J = 0$: $+$ ee, $-$ oo, 0 odd- A .

Validity. Best for $A \gtrsim 20$ away from closed shells; misses magic-number (N or $Z = 2, 8, 20, 28, 50, 82, 126$) and deformation effects. Maximum at $A \simeq 56$ (Fe), $B/A \simeq 8.8 \text{ MeV}$: fusion of light, fission of heavy, both release energy.

(b) Energy released per fission

$$\Delta E \simeq B(^{92}\text{Kr}) + B(^{141}\text{Ba}) - B(^{235}\text{U}).$$

Read B/A from the curve:

$$B(^{235}\text{U}) \simeq 235 \times 7.6 \text{ MeV} = 1786 \text{ MeV}.$$

$$B(^{92}\text{Kr}) \simeq 92 \times 8.7 \text{ MeV} = 800 \text{ MeV}.$$

$$B(^{141}\text{Ba}) \simeq 141 \times 8.4 \text{ MeV} = 1184 \text{ MeV}.$$

Sum (free neutrons unbound):

$$\Delta E = 800 + 1184 - 1786 \text{ MeV}.$$

$$\boxed{\Delta E \simeq 200 \text{ MeV per fission.}} \quad (15)$$

(c) Anti-neutrino rate from a 10 GW_{th} reactor

Each fission $\Rightarrow$ two neutron-rich daughters $\Rightarrow \sim 3 \beta^-$ each $\Rightarrow \bar{n}_\nu \simeq 6 \bar{\nu}_e/\text{fission}$. Fission rate:

$$R_{\text{fiss}} = \frac{P_{\text{th}}}{\Delta E} = \frac{10^{10} \text{ J/s}}{200 \text{ MeV} \times 1.602 \times 10^{-13} \text{ J/MeV}}.$$

Evaluate:

$$R_{\text{fiss}} \simeq 3.12 \times 10^{20} \text{ s}^{-1}.$$

Anti-neutrino rate:

$$R_{\bar{\nu}} = \bar{n}_\nu R_{\text{fiss}} = 6 \times 3.12 \times 10^{20} \text{ s}^{-1}.$$

$$\boxed{R_{\bar{\nu}} \simeq 1.9 \times 10^{21} \bar{\nu}_e/\text{s.}} \quad (16)$$

(d) Detector mass at 200 km for 10 events/month

Isotropic flux at $D = 200 \text{ km} = 2 \times 10^7 \text{ cm}$:

$$\Phi = \frac{R_{\bar{\nu}}}{4\pi D^2} = \frac{1.9 \times 10^{21}}{4\pi(2 \times 10^7)^2} \text{ cm}^{-2} \text{ s}^{-1}.$$

Evaluate:

$$\Phi = \frac{1.9 \times 10^{21}}{5.03 \times 10^{15}} = 3.78 \times 10^5 \text{ cm}^{-2} \text{ s}^{-1}.$$

Above-threshold fraction 10%:

$$\Phi_{\text{eff}} = 0.10 \Phi = 3.78 \times 10^4 \text{ cm}^{-2} \text{ s}^{-1}.$$

Free-proton density per gram of C₂₀H₃₄ ($M = 20(12) + 34 = 274 \text{ g/mol}$):

$$n_p = \frac{34 N_A}{M} = \frac{34(6.02 \times 10^{23})}{274} \text{ g}^{-1}.$$

Evaluate:

$$n_p \simeq 7.47 \times 10^{22} \text{ g}^{-1}.$$

Event rate per gram:

$$r = n_p \sigma_{\text{IBD}} \Phi_{\text{eff}} = (7.47 \times 10^{22})(4 \times 10^{-42})(3.78 \times 10^4) \text{ s}^{-1} \text{ g}^{-1}.$$

Evaluate:

$$r \simeq 1.13 \times 10^{-14} \text{ s}^{-1} \text{ g}^{-1}.$$

Counts per gram per month ($t = 2.59 \times 10^6 \text{ s}$):

$$N_1 = r t = 1.13 \times 10^{-14} \times 2.59 \times 10^6 \simeq 2.93 \times 10^{-8} \text{ events/g.}$$

Required mass for 10 events:

$$M_{\text{det}} = \frac{10}{N_1} = \frac{10}{2.93 \times 10^{-8}} \text{ g} \simeq 3.4 \times 10^8 \text{ g}.$$

$$\boxed{M_{\text{det}} \simeq 340 \text{ tonnes } (\sim \text{kT scale, KamLAND-class}).} \quad (17)$$

(e) Missing assumption

Above calculation assumes flavour conservation in propagation. *Neutrino oscillations* convert $\bar{\nu}_e \rightarrow \bar{\nu}_{\mu,\tau}$ over $L = 200$ km; IBD is sensitive only to $\bar{\nu}_e$. Survival

$$P_{ee} \simeq 1 - \frac{1}{2} \sin^2 2\theta_{12} \simeq 0.55,$$

in the KamLAND averaged regime ($\Delta m_{21}^2 L/4E \gtrsim 1$ for $E \sim$ few MeV at $L \sim 200$ km). Multiplies the predicted rate by ~ 0.5 .

Missing assumption: neutrino oscillations.
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(18)